Possible Questions / Tasks in the exam (list might not be comprehensive):

You should be able to:

(1) draw a sketch of the different BUG algorithms for simple environments.

(2) explain the concept of configuration space

(3) define sets like configuration space obstacles CO_i and freespace F

(4) draw a sketch of configuration space obstacles for a simple moving system (robot) with translational and/or rotational degrees of freedom.

(5) formulate the representation for different robot types in configuration space.

(6) construct configuration space obstacles for translational degrees of freedom and know the underlying concepts.

(7) explain the potential field method and its pros and cons

(8) explain and draw a visibility graph and a reduced visibility graph for a simple setup of obstacles.

(9) draw the generalized Voronoi diagram (GVD) for a very simple environment

(10) explain the **probabilistic roadmap** as a motion planning method and its characteristics ansd construction including the role of the local planner.

(11) explain the core idea of the **PRM variants** and what is their benefit.

(12) explain a **RRT and outline several steps** in constructing a RRT and be able to discuss variants of it.

(13) outline several steps in constructing a visibility roadmap (note: <u>not</u> a visibility graph) for a configuration space with a simple environment.

(14) explain how pseudorandom numbers are generated.

(15) explain metrics in quasirandom sampling.

(16) explain the concept of exapnsiveness and to discuss its pros and cons.

(17) explain how to deal with for moving objects or multirobot systems.